

The Goalkeeper in Possession

Supplementary Methods and Results v5.0

Jeffrey Eyestone | Peace FC Analytics

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S1. Full event-data cohort and model specification

The supplement preserves the analyses that validate the core event-data framework but are not required to carry the main-paper narrative. It includes portability/calibration, intended-target selection, downstream criterion validation, completion, receiver evidence, falsified historical claims, and detailed 360 selection/robustness diagnostics.

S2. Leave-one-competition-out portability

Held-out cohort	AUC	Brier	Calibration slope	Prediction gap
PL1516	0.645	0.217	0.673	+7.4 pp
T2024	0.681	0.170	1.019	-6.9 pp
WC2018	0.710	0.204	1.455	+6.2 pp
WC2022	0.683	0.163	1.301	-8.0 pp
WSL2021	0.645	0.171	0.996	-9.5 pp

Discrimination transfers moderately, but absolute probability scale does not. These results motivate local/environment calibration rather than direct transplantation of an xR probability from one football environment to another.

S3. Intended-target geometry and selection

Cohort	Selected n	TF AUC	TA-GEO AUC	Delta	95% CI
PL1516	19169	0.645	0.796	+0.151	+0.141 to +0.161
T2024	4921	0.666	0.910	+0.244	+0.224 to +0.268
WC2018	2716	0.707	0.859	+0.152	+0.129 to +0.174
WC2022	3701	0.681	0.903	+0.222	+0.198 to +0.249
WSL2021	9647	0.667	0.890	+0.223	+0.205 to +0.241
Cohort	Target ID retained	Target ID failed	TF AUC full	TF AUC target-ID	
PL1516	95.8%	61.0%	0.647	0.645	
T2024	99.2%	67.0%	0.672	0.666	
WC2018	99.4%	77.4%	0.703	0.707	
WC2022	98.0%	56.7%	0.676	0.681	
WSL2021	98.1%	63.4%	0.674	0.667	

Intended geometry is highly informative, but target identification is outcome-dependent. This is why the 360 extension is important: it measures visible teammate geometry at the event state without requiring a successful receiver outcome.

S4. Downstream criterion validation and ablation

Panel	Metric	Positive cells	Median rho	P10	Min	Max
keeper-level	B2_XR	15	0.210	-0.099	-0.129	0.709
action-decile	B2_XR	17	0.455	nan	-0.455	0.952
keeper-level	B3_SUCCESS_ONLY	11	0.043	-0.549	-0.683	0.674
action-decile	B3_SUCCESS_ONLY	11	0.067	nan	-0.830	0.721
keeper-level	B4_FIXED_FAILURE	11	0.026	-0.262	-0.421	0.759
action-decile	B4_FIXED_FAILURE	14	0.158	nan	-0.636	0.879
keeper-level	B5_COARSE_ORIGIN_FAILURE	10	0.007	-0.261	-0.410	0.759

Panel	Metric	Positive cells	Median rho	P10	Min	Max
action-decile	B5_COARSE_ORIG IN_FAILURE	15	0.194	nan	-0.612	0.915
keeper-level	B6_FULL_XTGTK	10	0.006	-0.276	-0.422	0.759
action-decile	B6_FULL_XTGTK	16	0.279	nan	-0.600	0.867
Comparison		Full better	Baseline better	No clear diff		Median delta rho
B6_vs_B2_XR		0	8	12		0.002
B6_vs_B3_SUCCESS_ONLY		7	0	13		0.153
B6_vs_B4_FIXED_FAILURE		0	0	20		0.000
B6_vs_B5_COARSE_ORIG IN_FAILURE		0	0	20		-0.001

The full value construction is not supported as a superior downstream predictor. xR is more stable. Failure pricing improves success-only value in some settings, while action-specific failure-cost detail does not clearly outperform simple failure penalties.

S5. Failure-asymmetry spatial gradient

Cohort	Origin band	n	C/V	95% CI	(V+C)/V
PL1516	0-16.5m	20102	1.44x	1.44-1.45	2.44x
PL1516	16.5-35m	2508	1.10x	1.09-1.11	2.10x
PL1516	35-52.5m	213	1.05x	1.03-1.08	2.05x
T2024	0-16.5m	4507	1.49x	1.47-1.51	2.49x
T2024	16.5-35m	816	1.36x	1.34-1.39	2.36x
T2024	35-52.5m	42	1.20x	1.13-1.30	2.20x
WC2018	0-16.5m	2474	2.04x	2.00-2.09	3.04x
WC2018	16.5-35m	464	1.39x	1.34-1.43	2.39x
WC2018	35-52.5m	16	1.28x	1.15-1.42	2.28x
WC2022	0-16.5m	3552	1.48x	1.45-1.51	2.48x
WC2022	16.5-35m	579	1.44x	1.39-1.49	2.44x
WC2022	35-52.5m	22	1.19x	1.12-1.29	2.19x
WSL2021	0-16.5m	8531	1.93x	1.91-1.95	2.93x
WSL2021	16.5-35m	2033	1.20x	1.17-1.22	2.20x
WSL2021	35-52.5m	85	0.96x	0.93-0.99	1.96x

S6. StatsBomb 360 coverage, model attribution, and robustness

The 360 extension uses 4,831 event-linked goalkeeper distributions across 115 matches, with 4,802 exact UUID joins to frozen xR. Covered versus uncovered samples differ in goal-kick and event-pressure prevalence; findings therefore apply to the covered sample. A verified raw-source manifest contains 348 nonzero official JSON files, and a clean rerun reproduces all 4,831 distributions and 27,777 candidate options field for field.

Scope	Metric	M2	M3	Delta	95% CI
ALL	AUC	0.6485	0.6991	+0.0506	+0.0348 to +0.0660
ALL	Brier	0.1661	0.1596	-0.0065	-0.0087 to -0.0042
ALL	log_loss	0.5046	0.4879	-0.0167	-0.0221 to -0.0111
WC2022	AUC	0.6371	0.6919	+0.0548	+0.0335 to +0.0763
WC2022	Brier	0.1665	0.1610	-0.0055	-0.0086 to -0.0026
WC2022	log_loss	0.5030	0.4893	-0.0137	-0.0215 to -0.0064
EURO2024	AUC	0.6339	0.6940	+0.0601	+0.0353 to +0.0829
EURO2024	Brier	0.1673	0.1592	-0.0081	-0.0118 to -0.0044
EURO2024	log_loss	0.5096	0.4892	-0.0204	-0.0289 to -0.0112

M2 is logit(frozen xR) + event pressure + goal-kick state + competition (competition omitted inside single-tournament models). M3 differs only by clear-short-option count and visible central short exit. The same match-grouped folds and action IDs are used for every nested model.

Scope	Camera-control metric	M3C-M2C delta	95% CI
ALL	AUC	+0.0528	+0.0367 to +0.0687
ALL	Brier	-0.0065	-0.0086 to -0.0043
ALL	log_loss	-0.0166	-0.0218 to -0.0113
WC2022	AUC	+0.0579	+0.0360 to +0.0798
WC2022	Brier	-0.0057	-0.0087 to -0.0027
WC2022	log_loss	-0.0146	-0.0225 to -0.0071
EURO2024	AUC	+0.0475	+0.0257 to +0.0692

Scope	Camera-control metric	M3C-M2C delta	95% CI
EURO2024	Brier	-0.0074	-0.0107 to -0.0040
EURO2024	log_loss	-0.0183	-0.0261 to -0.0100

S7. Coach-facing descriptive examples

Keeper	Cohort	State	n	Short	Medium	Direct	Retention	xR
Kasper Schmeichel	PL1516	FREE	1168	9.5%	10.9%	79.6%	53.6%	63.1%
Kasper Schmeichel	PL1516	PRESSURED	47	8.1%	8.1%	83.8%	57.4%	49.4%
Jordan Pickford	T2024	FREE	253	52.8%	16.3%	30.9%	79.4%	80.3%
Jordan Pickford	T2024	PRESSURED	51	34.0%	10.0%	56.0%	60.8%	65.1%
Grace Moloney	WSL2021	FREE	853	61.6%	16.9%	21.5%	81.5%	78.6%
Grace Moloney	WSL2021	PRESSURED	60	57.4%	18.5%	24.1%	75.0%	60.3%

These profiles are descriptive examples for video review and match planning. They should not be interpreted as causal pressure responses or as universal player rankings.

S8. Claims falsified or downgraded during development

Earlier claim	Status	Final interpretation
Completion and value never agree	Rejected	Completion is positively associated with value; it is informative but insufficient.
xT-GK is style-neutral	Rejected	Final value remains associated with distribution style.
Zero-shot probability calibration transfers almost perfectly	Rejected	Unseen-environment gaps reach roughly 6-10 percentage points.
40-60 m goal kicks are universally most costly	Rejected	Corrected units/target analysis do not reproduce a universal shape.
Full xT-GK is a superior downstream predictor	Rejected	xR is more stable; full xT-GK is a consequence scale rather than a superior prediction metric.
Action-specific failure cost clearly beats simpler failure penalties	Rejected	No clear incremental advantage in public event data.
Transported C/V ~2.2 is a universal magnitude	Downgraded	Independent cohort refits produce C/V 1.40-1.90 while preserving the directional asymmetry.

S9. Reproducibility provenance

The public reproducibility repository contains the event-model source, match manifests, model-selection locks, OOF result tables, 360 source manifest and pinned upstream commit, 360 option-table generator, R1.2B nested-model scripts, environment lock, figure-generation scripts, and verification commands. Raw public provider files are retrieved from documented official URLs rather than bundled when licensing or package size makes rehosting undesirable. Repository URL: <https://github.com/JeffreyEyestone/goalkeeper-in-possession>.